

Market Survey:

Catching The PULSE Of MEDICAL ELECTRONICS MARKET



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The strong growth of the Indian economy also provides a boost to the country's healthcare industry. This, in turn, creates a huge demand for electronic components, devices and products that are used for medical purposes. Medical electronics technology reinforces the existing healthcare infrastructure in various ways, right from digitising medical tests, diagnostic and therapeutic procedures, to enhancing the reach of healthcare through telemedicine and health information technology.

A rise in the number of hospitals and an increased requirement for healthcare facilities create a need for sophisticated devices and equipment, which can provide accurate treatment to individuals. Medical electronics equipment/devices enable control, conversion, sensing, processing, storage, display and transfer of information on human anatomy and physiology by making use of electronics and communication technologies.

This discourse will provide an overview of the Indian medical electronics market with respect to size, growth drivers, major segments, trends, opportunities and challenges.

There are varied understandings of what constitute medical electronics. In this article, medical electronics refers to all electronic devices and equipment used in the diagnosis,

treatment and monitoring of patients. It does not include surgical devices, consumables like gloves, reagents and non-electronic implants used in treatments like orthopaedic and dental.

Market opportunity

Use of electronic equipment in the medical field for clinical and research purposes has been enhanced tremendously in the past few years. Human body generates numerous signals that are highly significant for diagnosis and therapy. These signals are collected from the body surface or from within, through electrodes of different sizes, shapes and types.

Moreover, there are non-electrical body parameters like temperature, blood flow, blood pressure and respiratory functions that need to be routinely monitored. These parameters are converted into corresponding electric signals by various transducers, and here comes the role of electronics. Extensive use of computers and microprocessors is now introduced in medical instruments designed to perform routine clinical measurements, particularly in those situations where data computing and processing could be considered as part of measurement and diagnostic procedures.

The medical electronics industry has achieved double-digit growth in recent years, and this growth trajectory is expected to continue owing to the rising incidence of chronic diseases, increased urbanisation and a growing elderly population and increased awareness about latest technology solutions to save, augment life and rehabilitation.

According to a Deloitte report, the Indian medical electronics device industry has been growing at an average rate of 17 per cent for the past few years and is expected to reach close to US\$ 6.5 billion by the year 2020 (Fig. 1).

Semiconductors play an important role in shaping the medical electronics device market. That market, in turn, may represent a significant growth opportunity for semiconductor players.

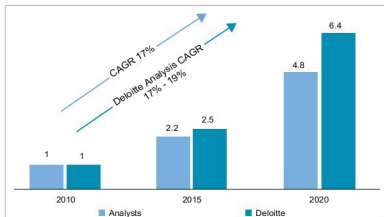


Fig. 1: Indian medical electronics market (in US\$ billion) (Source: FICCI-Deloitte report)